

and 68% vs. 79% and 65%, respectively ($p=0.01$). The 5-, 10-, and 15-year OS and CSS rates of all patients were 83%, 67% and 50%, and 97%, 95%, and 95%, respectively.

Conclusion: To the best of our knowledge, this is the largest database analysis of MAC. This study demonstrated that MAC is mainly a locally invasive disease with rare metastasis to lymph nodes or distantly. MAC has a long course of disease with favorable OS and CSS. The primary management remains surgical resection, though this study suggests PORT may improve outcomes in high risk patients.

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Exploiting Molecular Subtype Cell Plasticity As Novel Strategy for Targeting Glioma Stem Cells Through Alternating Therapy



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Purpose/Objective(s): Glioblastoma (GBM) is the most commonly diagnosed glioma and has poor median survival (14 months) despite aggressive trimodal therapy, which includes resection, radiation therapy (RT) and Temozolomide (TMZ). Recent genome-wide expression profiling has separated GBM into several distinct subtypes including proneural (PN) and mesenchymal (MES). PN and MES brain tumor initiating cells (BTICs) have mutually exclusive gene signatures and opposed sensitivities towards specific therapies. Interestingly, radiation triggers a shift from PN toward MES subtype, which is accompanied by increased sensitivity towards TMZ. Similarly, we have reported that exposure of MES BTICs to TMZ results in a loss of MES features and gain in PN characteristics, including upregulation of Nestin and down-regulation of TGM2. In addition, MES BTICs treated with TMZ exhibited greater response to subsequent chemoradiation compared to upfront concurrent RT and TMZ. We hypothesize that therapy-induced switches in BTICs subtypes drives alternating vulnerabilities to components of standard care (TMZ/RT), which therefore should be applied in an alternating mode rather than concomitantly. Furthermore, we propose that such alternating therapeutic regimen should be designed based on tumor subtype, such that PN tumors should receive neoadjuvant RT while MES should be treated with neoadjuvant TMZ to match the expected predominant subtypes of BTICs

Materials/Methods: We utilised an orthotopic mouse model of GBM using a panel of patient-derived BTICs of PN and MES identity and the XRAD-225cx irradiation platform, which allows precise imaging and RT delivery to recapitulate clinically relevant protocols in a pre-clinical setting. Using this model, we have assessed a panel of BTIC xenografts for response to neo-adjuvant TMZ or RT protocols to identify predictors of response. Furthermore, we investigated the mechanism driving improved response from neo-adjuvant TMZ protocols.

Results: We demonstrate that PN and MES have preferential sensitivity to RT or TMZ, respectively. Exposure of MES BTICs to TMZ caused loss of MES features (TGM2) and a gain of PN features (Nestin, CD15), which was accompanied by an increase in radiation sensitivity. We demonstrate that neo-adjuvant TMZ is beneficial towards MES xenografts and is driven by an initial MES to PN shift, which improves subsequent response to chemoradiation. We also explore the clinical effectiveness of protocols alternating between therapies and the impact of RT versus TMZ upfront for MES and PN xenografts.

Conclusion: Neo-adjuvant TMZ is preferentially efficacious in MES tumors where initial exposure to TMZ drives evolution towards a PN-like status, which has a greater response to subsequent chemoradiation. Development of a personalized protocol based upon initial assessment of molecular subtype is warranted.

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Update of Comparison of Low-Dose and Standard-Dose Total Skin Electron Beam Therapy for Mycosis Fungoides



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Purpose/Objective(s): Total skin electron beam therapy (TSEBT) is has been shown to be an effective treatment for mycosis fungoides (MF). Doses of 30-36 Gy have traditionally been used with high response rates, although treatment is associated with toxicity and most patients ultimately progress. Recent studies have proposed 12 Gy as an effective and tolerable dose, and our institution has been utilizing this regimen to treat MF patients since 2015. We compare our experiences using low-dose TSEBT and standard-dose TSEBT.

Materials/Methods: Between 2010 and 2016, 26 patients with Stage IB – IVA MF were treated with TSEBT at our institution. Thirteen patients received a total of 16 courses of low-dose TSEBT to 12 Gy, with 3 of these patients received a repeat course of low-dose TSEBT during this time. As a comparison group, 13 patients received standard-dose TSEBT between 30-36 Gy, all from 2015 and prior. We compare response rates, time to progression and toxicity between these treatment groups.

Results: Median follow-up was 15.8 months. Patients who received low-dose TSEBT were more likely to have had prior local radiation treatment (50% versus 8%, $p = 0.013$). All patients experienced at least a partial response to treatment, with 31% of the low-dose TSEBT courses and 77% of the standard-dose TSEBT courses resulting in initial clinical complete response ($p = 0.014$). Patients who underwent a repeat course of low-dose TSEBT had a 67% complete response rate as compared to 13% with single course low-dose TSEBT and 77% with standard-dose TSEBT ($p = 0.020$). Fourteen of 16 courses of low-dose TSEBT and 12 of 13 courses of standard-dose TSEBT resulted in progression. The average time to progression for low-dose TSEBT was 3.2 months versus 8.2 months for standard-dose TSEBT ($p = 0.030$). Common toxicity to both treatment arms was pain and pruritus. Compared to low-dose TSEBT, standard-dose TSEBT trended toward increased toxicity with hyperpigmentation/erythema (23% versus 6%, $p = 0.19$) and dry desquamation (46% versus 19%, $p = 0.11$). Low-dose TSEBT trended towards higher rates of skin dryness compared to standard-dose TSEBT (56% versus 23%, $p = 0.071$). Rates of dry desquamation increased to 33% with repeat low-dose TSEBT compared to 15% with single course low-dose TSEBT and 46% with standard-dose TSEBT ($p = 0.24$).

Conclusion: Low-dose TSEBT was associated with a significantly lower complete response rate and shorter time to progression compared to standard-dose TSEBT, although rates of complete response improved with a repeat course of low-dose TSEBT. In addition, low-dose TSEBT trended toward lower rates of toxicity compared to standard-dose TSEBT. Given the high rate of progression regardless of treatment regimen, the use of low-dose TSEBT should be considered for the treatment of MF as this approach allows for re-treatment with complete response rates rivaling that of standard-dose TSEBT with a better toxicity profile.

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Ocular Brachytherapy for Intra-ocular Melanoma: Experience and Outcomes in The Province of Alberta



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Introduction: Uveal melanoma is the most common cause of intra-ocular malignancy, and accounts for about 5% of melanoma diagnoses. It has a 12 year overall survival approaching 60%, and disease specific mortality ranging from 17% to 21%. Treatment historically was only through